

MESSOU Franck

Generative AI / Backend / Software Engineer | PhD Researcher | Full-Stack, MLOps & Applied LLMs

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● Summary

PhD researcher in Applied Informatics (Hosei University, YuLab) and hands-on engineer spanning AI, full-stack, and MLOps. Research on LLMs / small language models and time-series methods for smart-grid stability and IoT systems (2 Q1 journal papers, 5 IEEE conference papers), alongside shipped products: a deployed web platform (cvflow.app), an AI meeting-assistant app (offline sLM + Gemini / GPT), and multi-GPU model-serving infrastructure. Comfortable from research and model training through REST APIs, cloud deployment (GCP), and cross-platform apps.

● Technical Skills

AI / ML & LLMs: PyTorch, Hugging Face Transformers, vLLM, scikit-learn, LLM / sLM fine-tuning (LoRA, PEFT), RAG; Gemini, GPT, Claude, Llama, Qwen, Phi

NLP, Speech & Methods: Speech-to-text, summarization, translation, time-series forecasting, anomaly detection, graph neural networks, distributed training (DDP)

Backend & APIs: Python, FastAPI, Flask, Node.js / Express, REST, GraphQL, JWT / OAuth2

Frontend & Mobile: Java, Flutter / Dart, React, HTML5, CSS3, TypeScript

Databases: PostgreSQL, MongoDB, MySQL, Firebase / Firestore, SQLite

Cloud, DevOps & MLOps: GCP (Cloud Run, Compute, Storage, IAM), Firebase, Docker, CI/CD (GitHub Actions), CUDA, Linux, model serving

● Experience

■ ML / Data Engineer

Oct 2025 – Jul 2026

D'isum Inc.

Tokyo, Japan

- Built ETL pipelines from raw sensor streams (50+ engines, 200+ batteries) to analytics-ready datasets, and integrated classical ML models for sound and time-series analysis into client systems.
- Built on-device iOS / Android sensor apps (Swift, Java) for real-time capture and edge inference.

■ Doctoral Researcher

Sep 2025 – Present

YuLab, Hosei University

Tokyo, Japan

- Designed dual-domain graph-neural-network architectures for power-system stability analysis on the PEGASE 2,869-bus grid.
- Built LLM / small-language-model fine-tuning pipelines (Spec2Llama, Spec2LLM, LLM4Imp) for time-series forecasting and imputation on edge devices, published at IEEE ICCE / ICC 2026.
- Provision and operate the lab's multi-GPU server: distributed training (DDP), checkpointing, and vLLM inference serving in Docker.

■ Master's Researcher

2023 – Sep 2025

YuLab, Hosei University

Tokyo, Japan

- Built transformer time-series forecasting models (TAPformer, TSFormer, CNN-BiLSTM); won the IEEE VTS 2024 Student Paper Award.

■ Generative AI Intern · TICAD9 Booth

Jul 2024 · Aug 2025

Japan AI Consulting (J-AIC) / JICA

Osaka · Yokohama

- Built GenAI prototypes for the consulting practice using Flask, PostgreSQL, and LLM APIs (Jul 2024 internship).
- Represented J-AIC and presented applied ML / GenAI prototypes at JICA's TICAD9 booth, Yokohama (Aug 2025).

■ Teaching & Research Assistant

2024 – Present

IIST, Hosei University

Tokyo, Japan

- Supported professors in English-language Project-Based Learning and graduation research for 4th-year undergraduates.

■ Full-Stack Developer

2022 – 2023

Anohou Solutions (anohou.dev)

Remote

- Built cross-platform mobile apps and backends for commercial products (OKOHI, Reçu ChapChap, EKKOU): Flutter / Dart clients, REST APIs, and MongoDB / PostgreSQL data layers.
- Delivered full-stack features end-to-end, from API design to mobile UI, in remote collaboration.

■ Software Engineer

Casys Technology (casys-technologies.net)

Dec 2019 – Jan 2023

Abidjan, Côte d'Ivoire

- Full-stack development of commercial utility applications used by clients across the ECOWAS region (Côte d'Ivoire, Benin, Mali, Togo, Burkina Faso, Guinea).
- Built and maintained web and backend services across the lifecycle: design, implementation, deployment, and support.

● Projects

■ Smart-grid co-optimization | PyTorch, graph neural networks, CUDA

2025 – Present

- Dual-domain GNN architectures for power-system stability at 2,869-bus scale (PhD research).

■ AIReco – AI meeting assistant | Flutter, GCP, on-device sLM, Gemini / GPT, ASR, NMT

2025 – 2026

- Records and transcribes meetings, then generates summaries, keywords, and translations across 20 languages.

■ CVFlow (cvflow.app) | Next.js, FastAPI, GCP, Firestore, Gemini, Stripe

2025 – Present

- AI resume builder: Gemini resume parsing, ATS scoring vs job descriptions, cover-letter and interview generators; 20-language UI; deployed on Cloud Run, built in remote international collaboration.

Public code: github.com/mesabo | github.com/hosei-university-iist-yulab

● Selected Publications

- **MFJ Aboya** et al., "TAPformer: Long-Term Electricity Load Forecasting in IoT Systems," *Internet of Things*, Elsevier, 2026. (IF 7.6, Q1)
- **MFJ Aboya** et al., "Spec2Llama: Spectral-Aware Forecasting for Versatile Time Series," *Expert Systems with Applications*, Elsevier, 2026. (IF 7.5, Q1)
- **MFJ Aboya** et al., "Hybrid Attention-Based CNN-BiLSTM for Short-Term Load Forecasting in IoT," *IEEE VTC2024-Fall*, 2024. (Student Paper Award)
- **MFJ Aboya** et al., "TSFormer: Temporal-Aware Transformer for Multi-Horizon Forecasting," *IEEE ICUFN 2025*.
- **MFJ Aboya** et al., "Spec2LLM: Spectral-to-Language Reprogramming for Power Transformer Temperature Forecasting," *IEEE ICCE 2026*.
- **MFJ Aboya** et al., "LLM4Imp: Frozen LLMs with Spectral Prompts for Time-Series Imputation," *IEEE ICC 2026*.
- **MFJ Aboya** et al., "Large Language Models for Temporal Sensor Networks: Graph-Constrained Multi-LLM Ensembles," *IEEE VTC2026-Spring*.

● Awards & Service

- IEEE VTS Tokyo / Japan Chapter Student Paper Award & Encouragement Award, VTC2024-Fall (2024).
- JICA Development Studies Program graduate, 2025.
- Review Committee, IEEE ICECET 2026 (Rome); Reviewer, *Theoretical Computer Science*, Elsevier.

● Education

■ PhD, Applied Informatics

Hosei University

Sep 2025 – Apr 2027 (early)

Tokyo, Japan

■ MSc, Applied Informatics

Hosei University

2023 – Sep 2025

Tokyo, Japan

● Languages

Spoken: French (native), English (fluent), Japanese (simple daily conversation, ongoing study)